

# Engineering Curves: Construction of Epicycloid

Rolling Circle Method with Tangent & Normal

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# Problem Statement

## Problem

A generating circle of diameter 30 mm rolls externally without slipping on a directing circle of diameter 90 mm. Construct one complete epicycloid arch corresponding to  $\angle AOB = 120^\circ$ , divide the arch into 12 equal angular parts, and retain every construction element in all subsequent views.

# Given Data and Calculations

## Given data

Diameter of directing circle = 90 mm,

$$R = 45 \text{ mm},$$

Diameter of generating circle = 30 mm,

$$r = 15 \text{ mm},$$

$$\angle AOB = 120^\circ,$$

Let number of divisions = 12.

## Calculations

$$OC_0 = R + r = 45 + 15 = 60 \text{ mm},$$

$$AC_0 = r = 15 \text{ mm},$$

$$\Delta\theta = \frac{120^\circ}{12} = 10^\circ.$$

## Generating-circle division

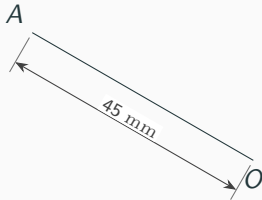
$$\text{Division angle} = \frac{360^\circ}{12} = 30^\circ.$$

## Rolling relation

Each  $10^\circ$  centre step  $\leftrightarrow$   $30^\circ$  generating-circle step.

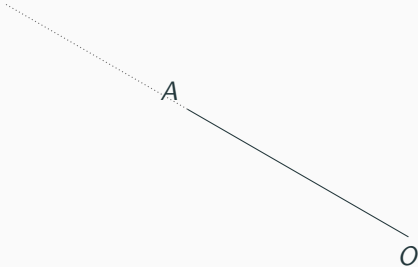
## Step 1: Draw $OA$

$OA = 45 \text{ mm.}$



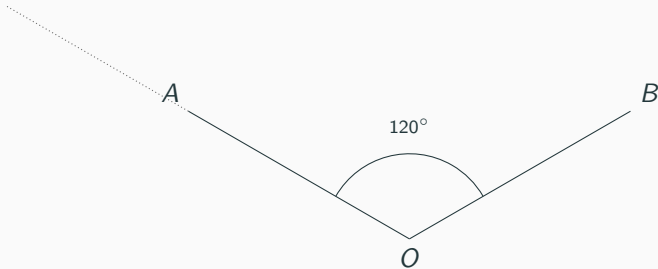
## Step 2: Extend $OA$

Extend  $OA$ .



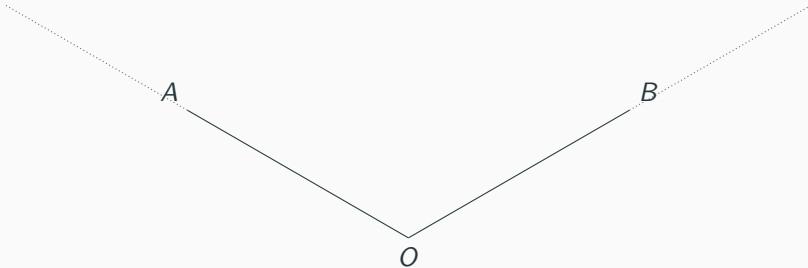
### Step 3: Draw $OB$ at $120^\circ$

$$\angle AOB = 120^\circ.$$



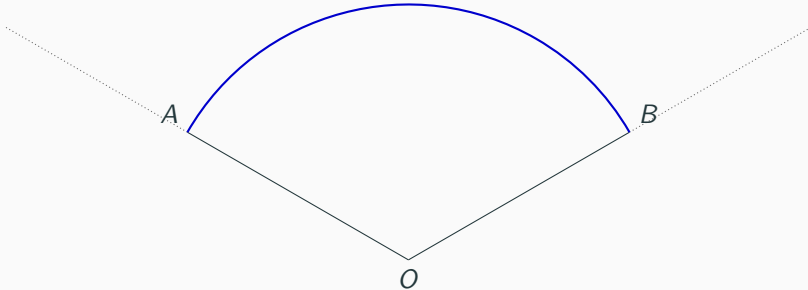
## Step 4: Extend $OB$

Extend  $OB$ .



## Step 5: Draw the directing arc $AB$

Blue arc  $AB$ : directing circle.

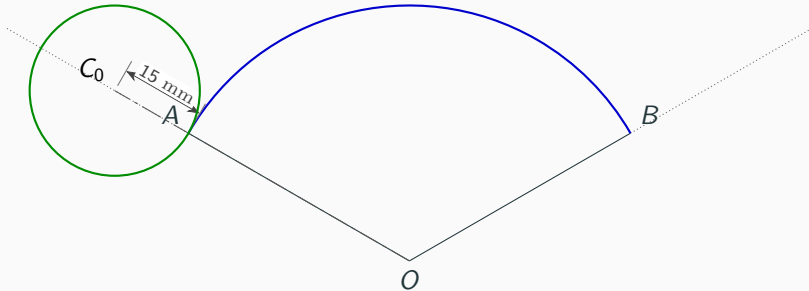




## Step 6: Locate $C_0$ and the generating circle

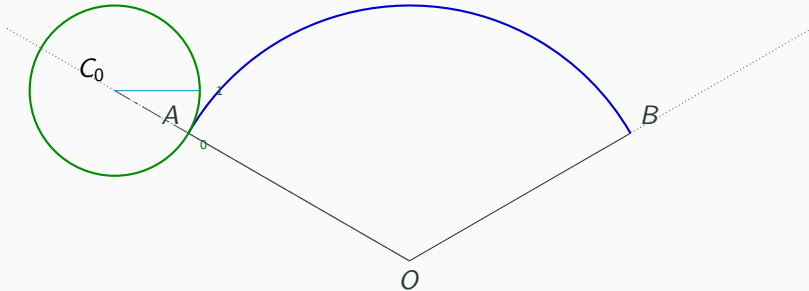
$AC_0 = 15 \text{ mm.}$

Green: generating circle.



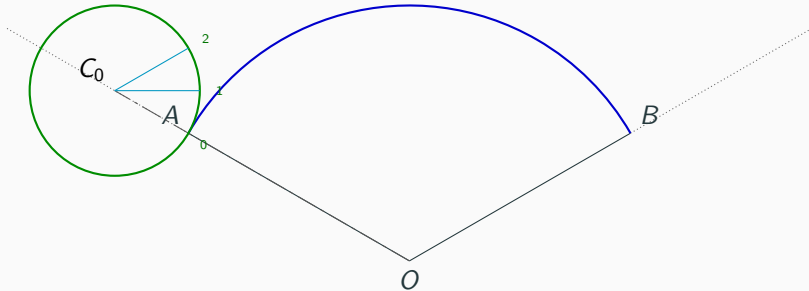
## Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.  
30° divisions.



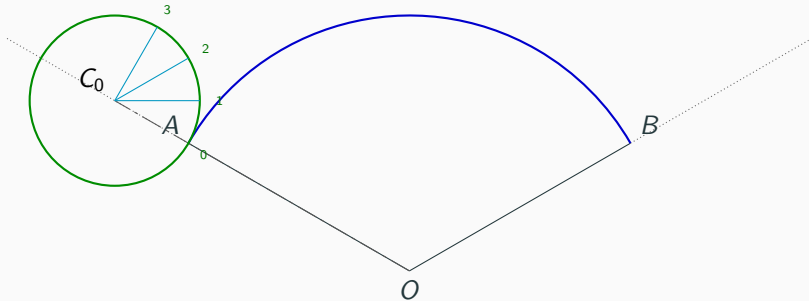
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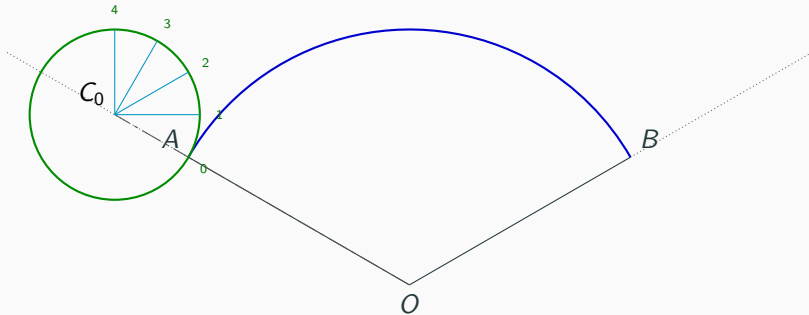
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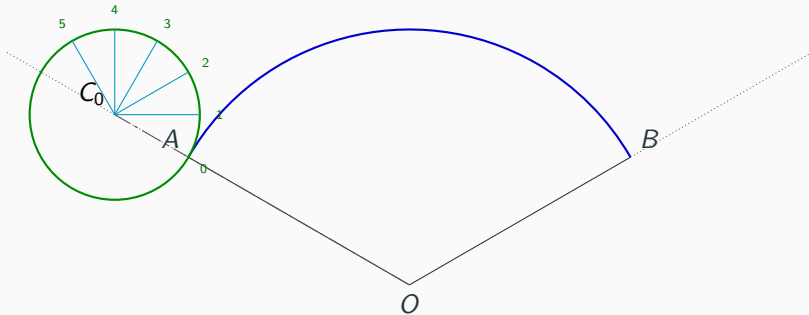
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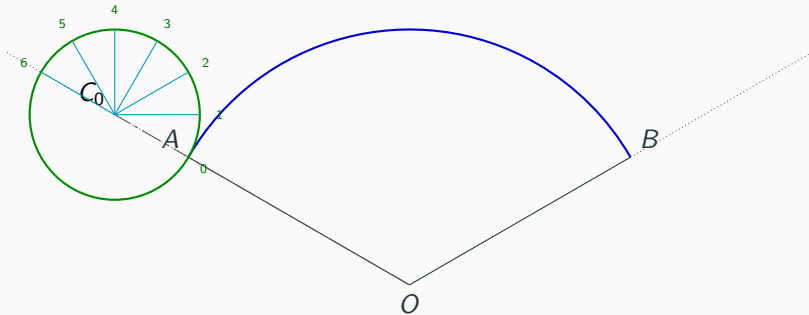
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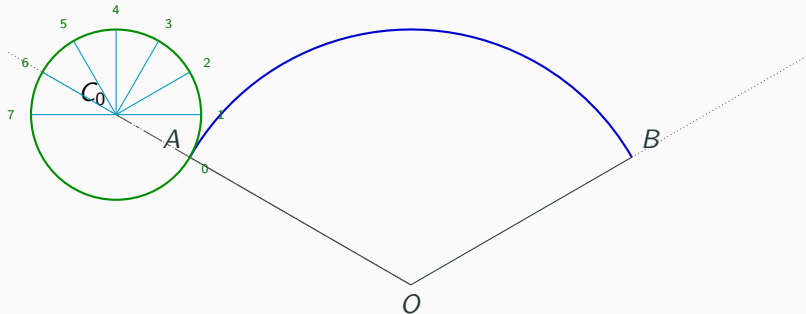
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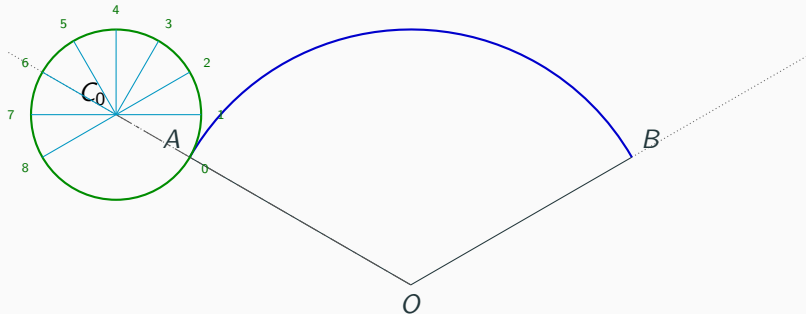
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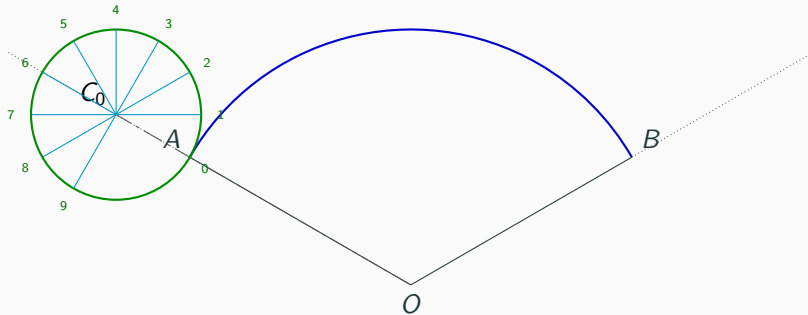
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30° divisions.



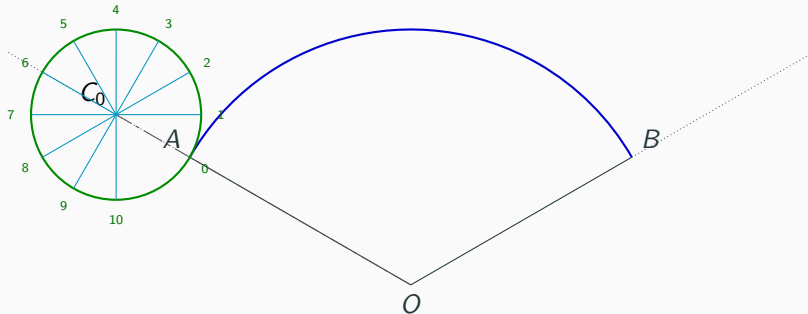
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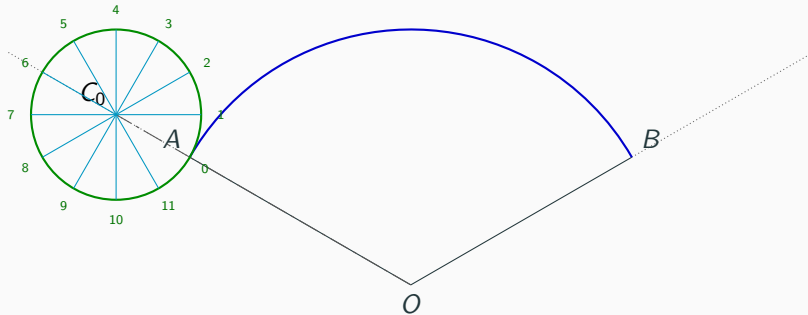
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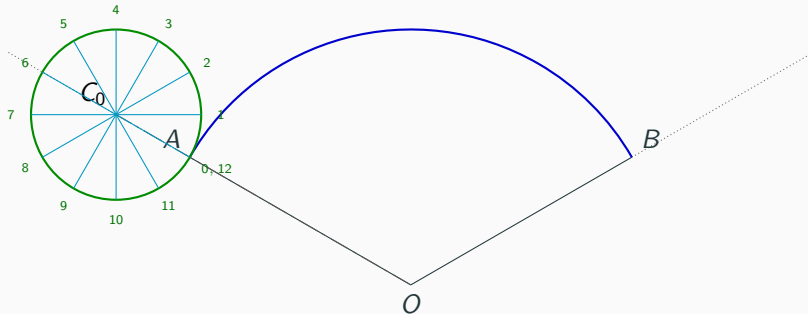
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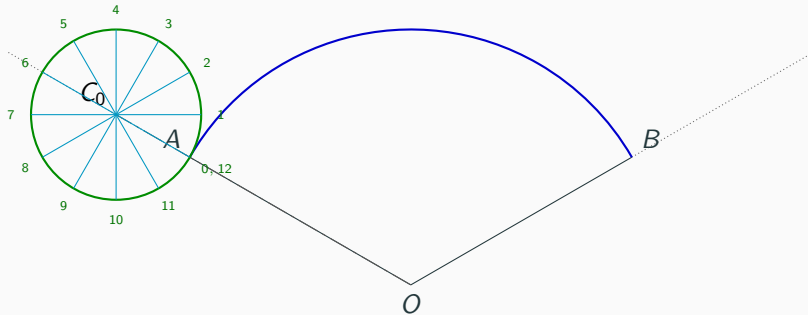
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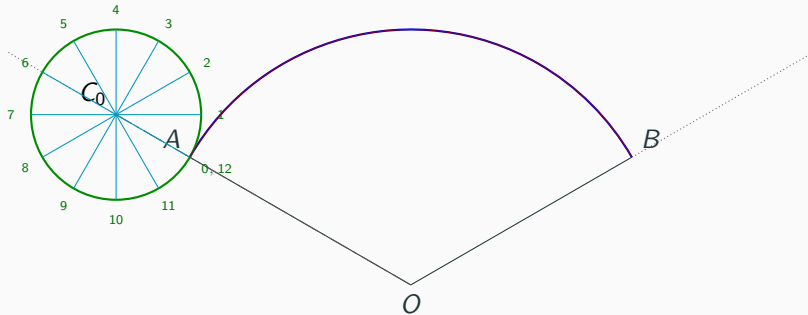
## Step 8: Complete the generating-circle divisions

Generating-circle divisions.



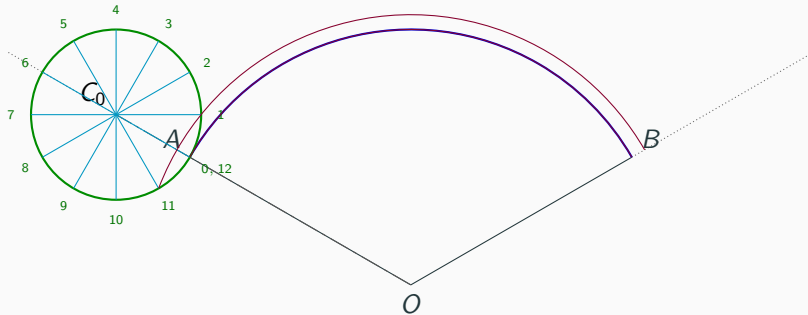
## Step 9: Draw concentric construction arcs

Concentric construction arc.



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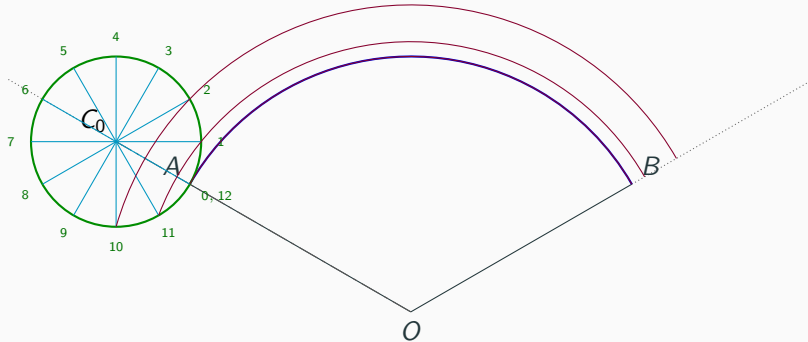
Concentric construction arc.





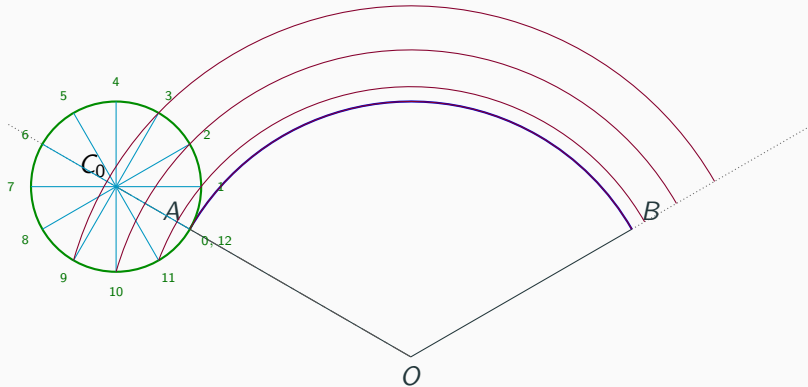
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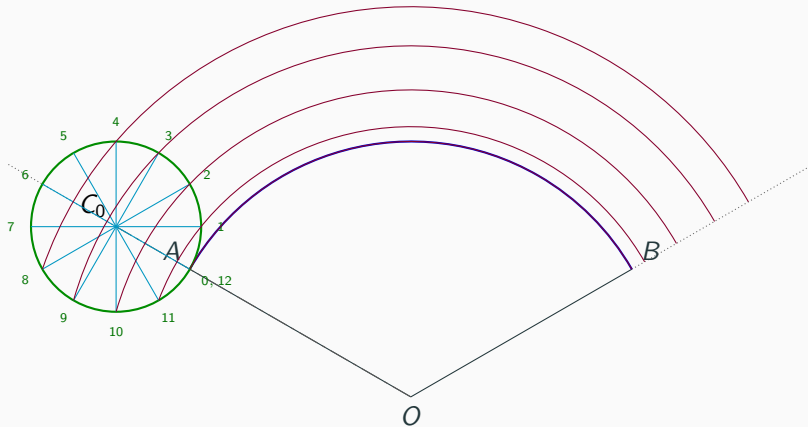
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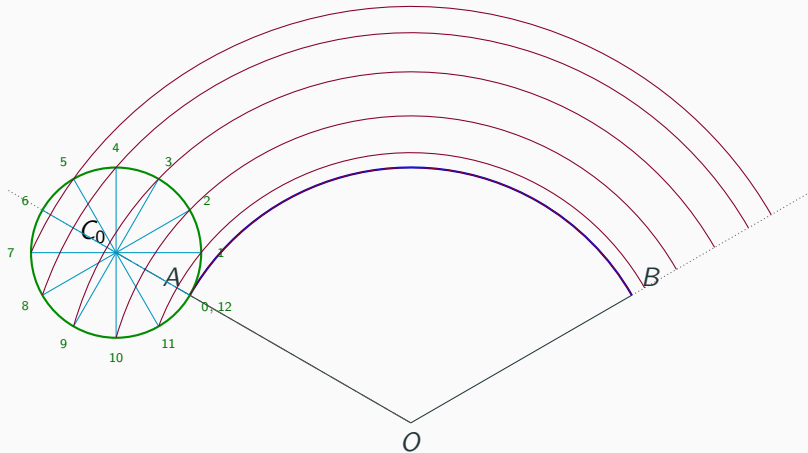
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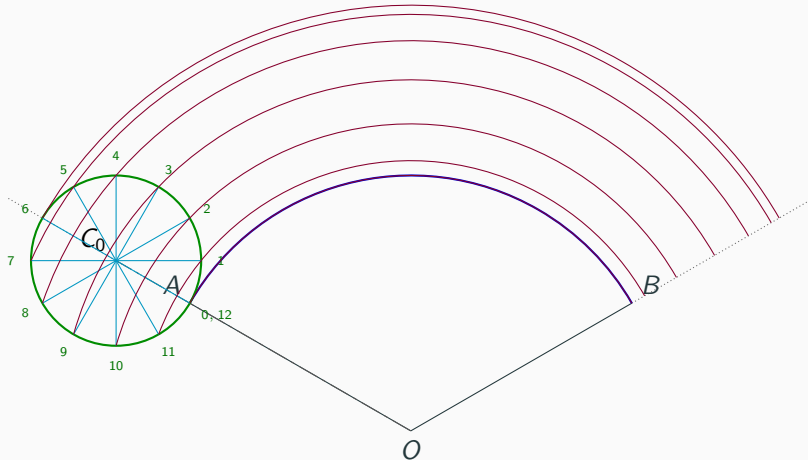
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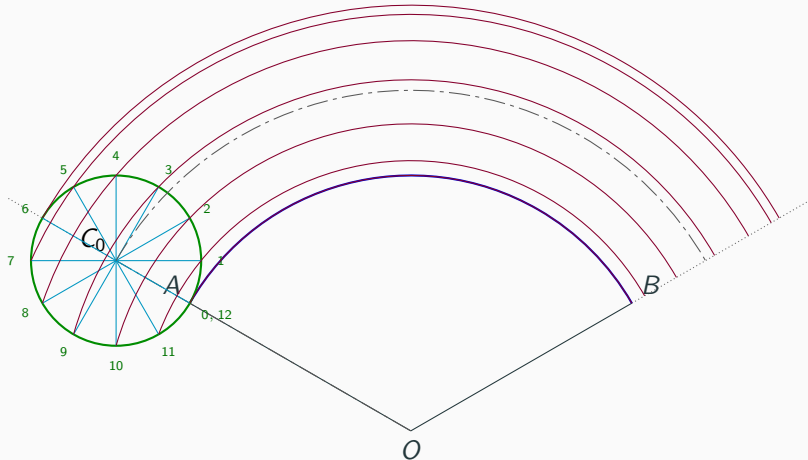
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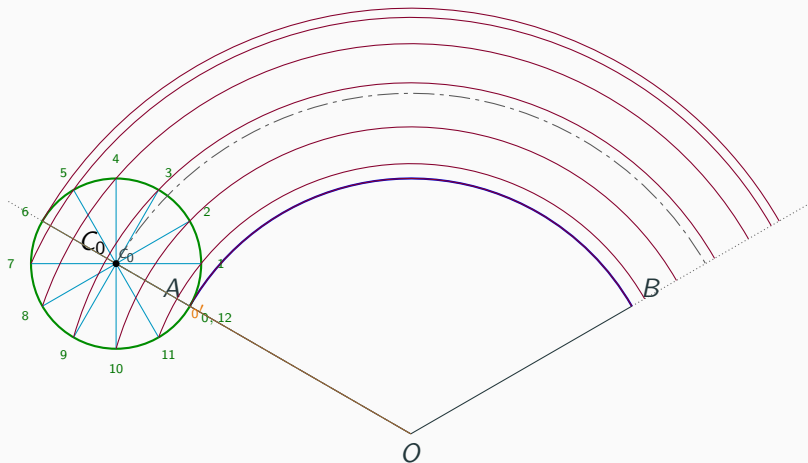
## Step 10: Draw the centre arc through $C_0$

Centre curve: ISO axis line.



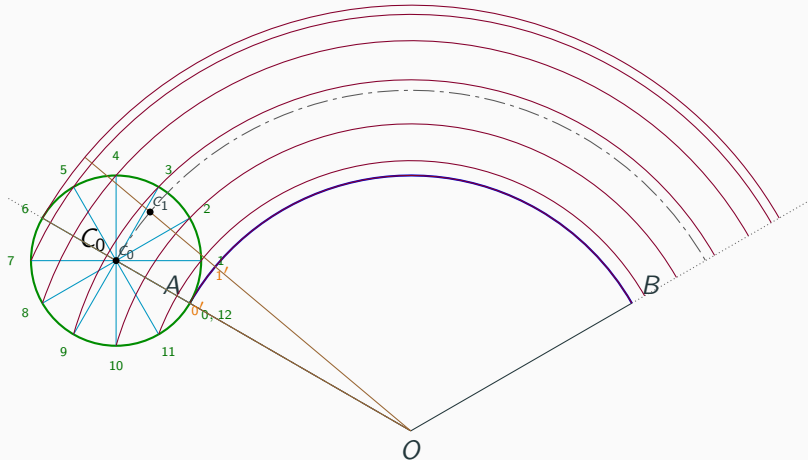
## Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



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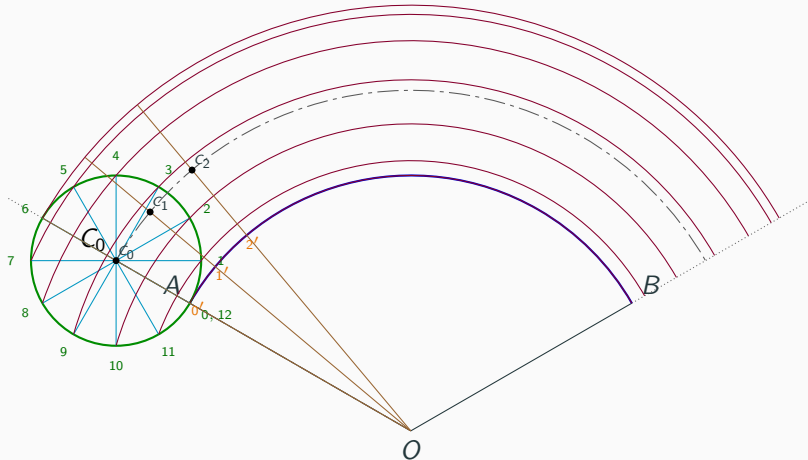
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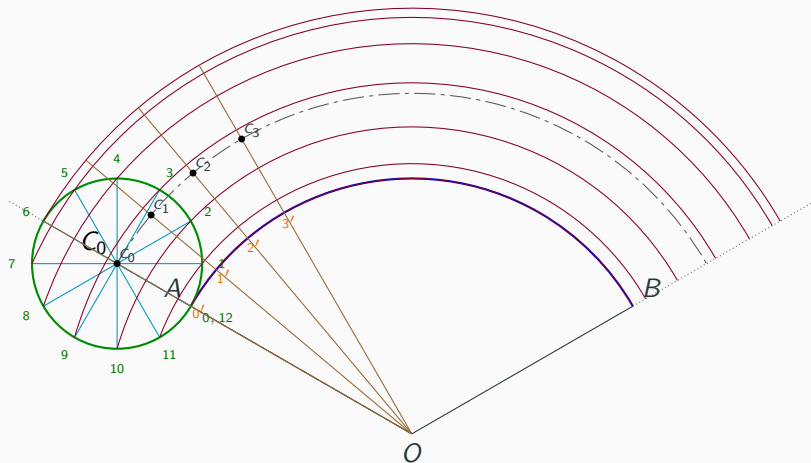
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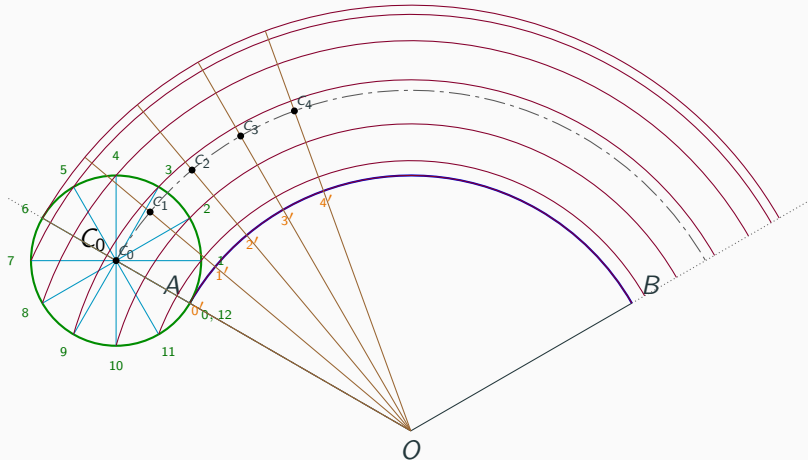
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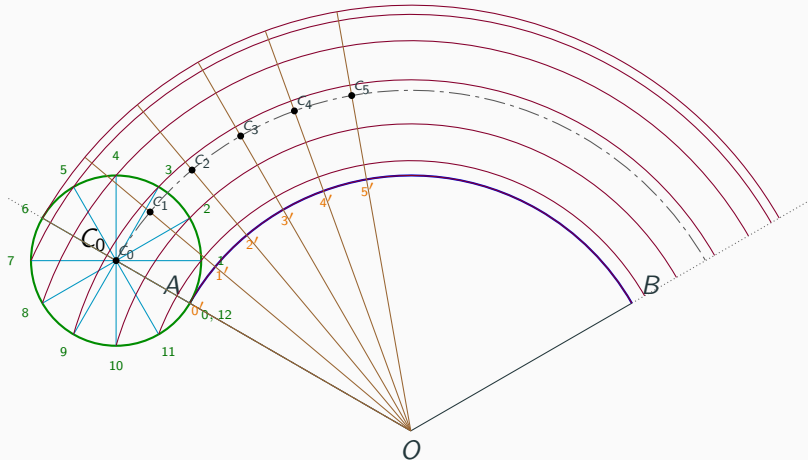
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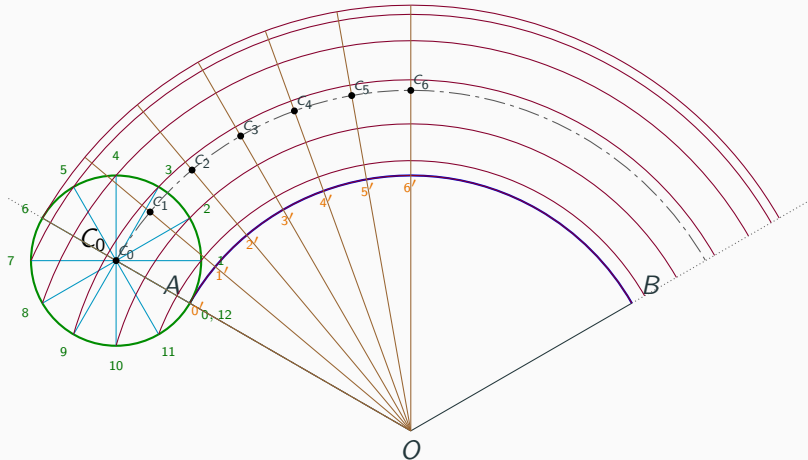
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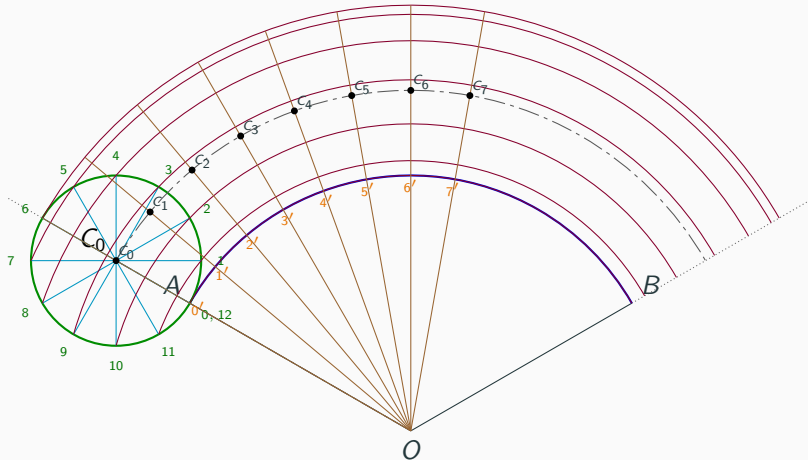
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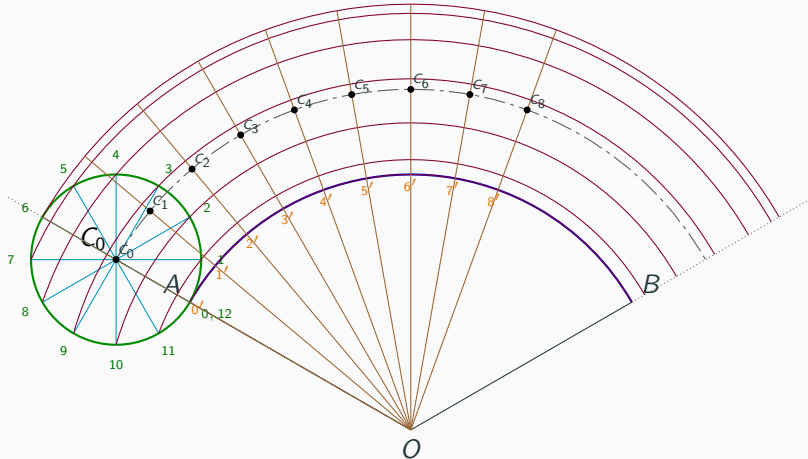
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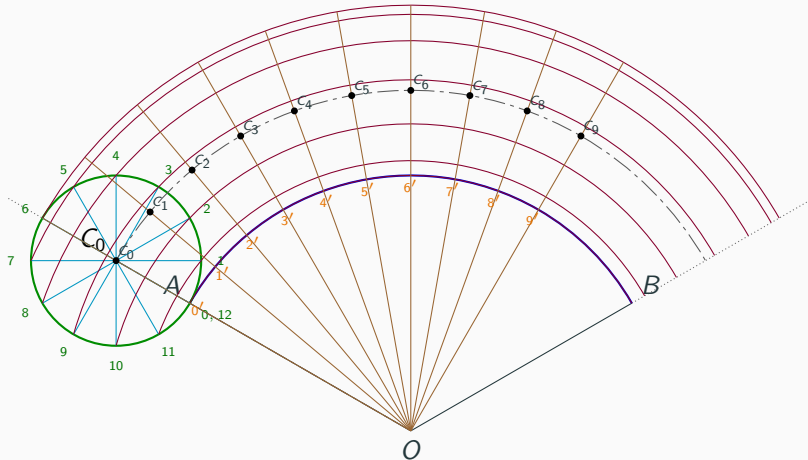
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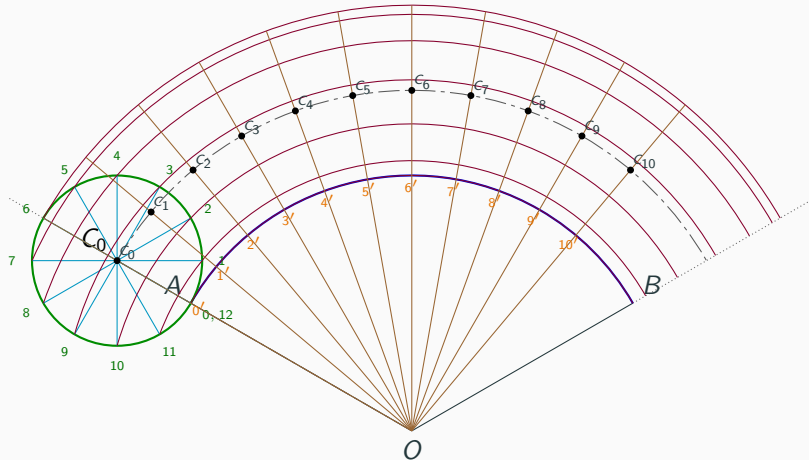
10° divisions.





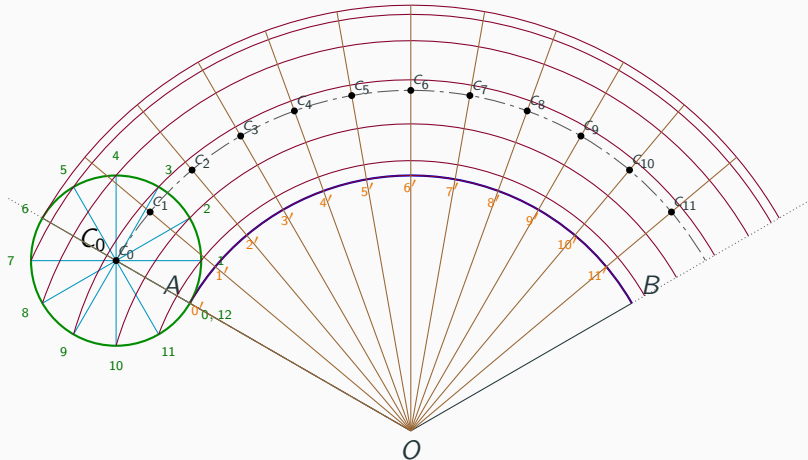
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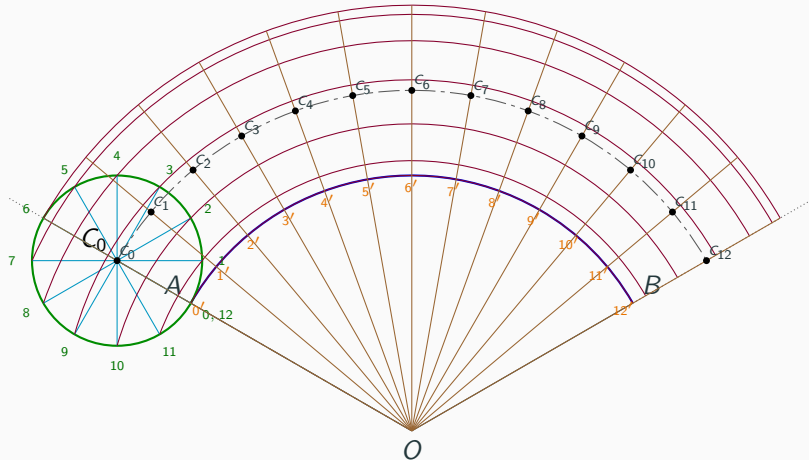
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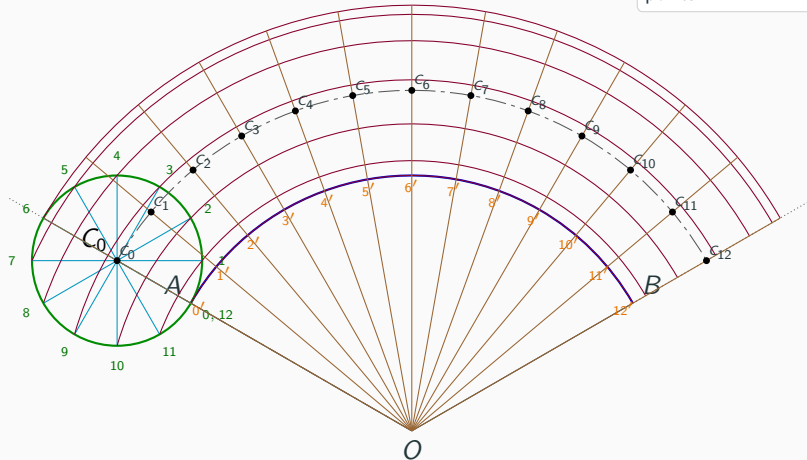
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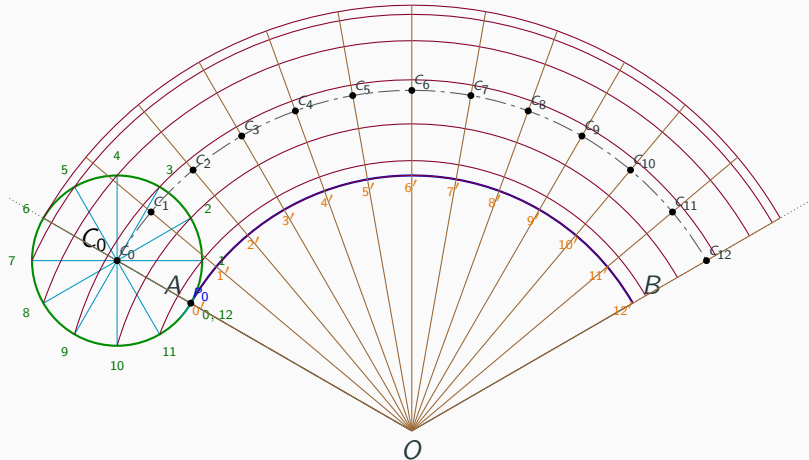
## Step 12: Complete the angular divisions and points

Division points and centre points.



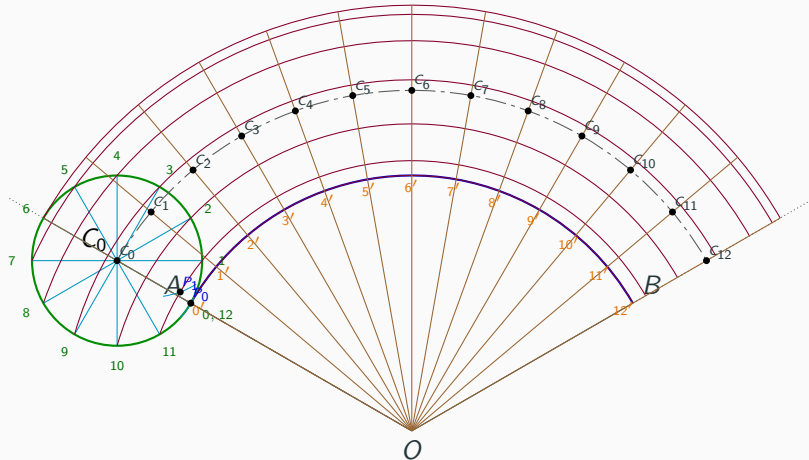
## Step 13: Draw the small cutting arc and locate $P_i$

Cyan arc: radius 15 mm.



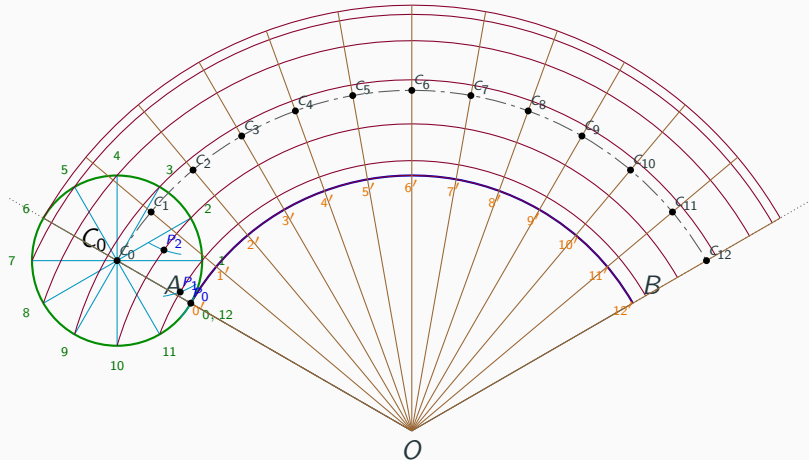
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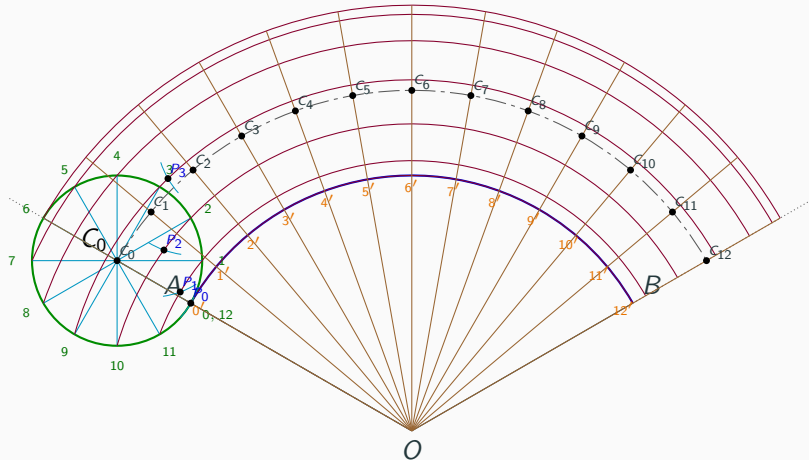
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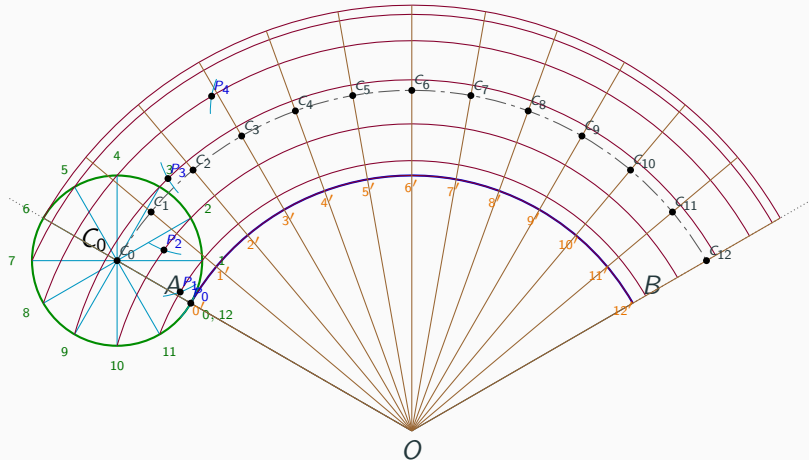
Cyan arc: radius 15 mm.





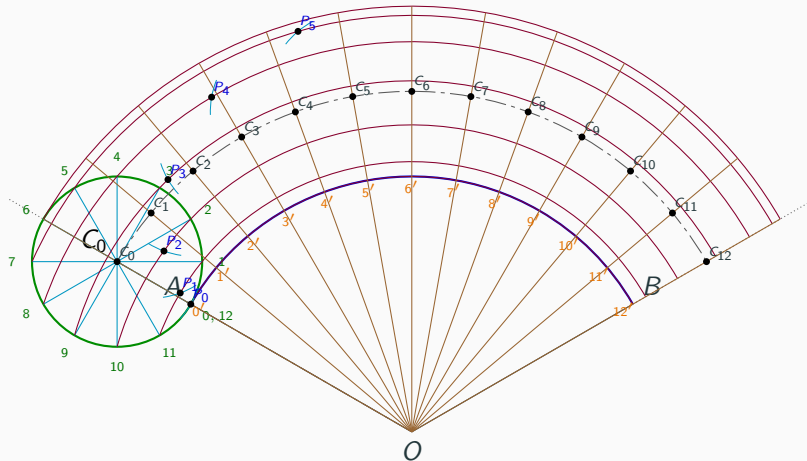
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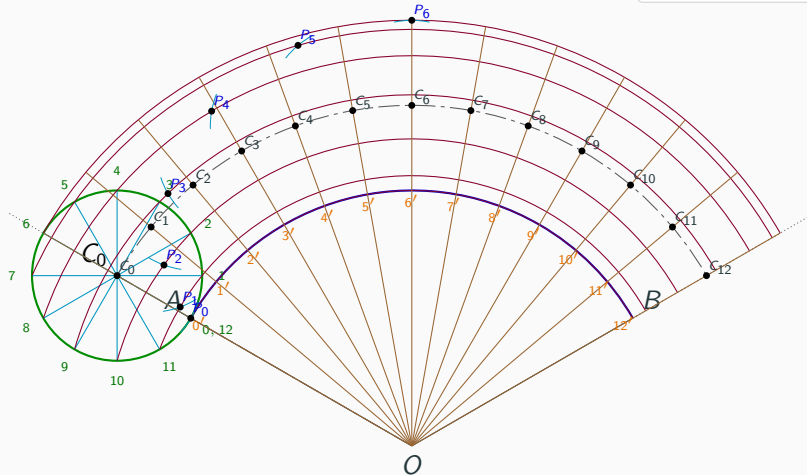


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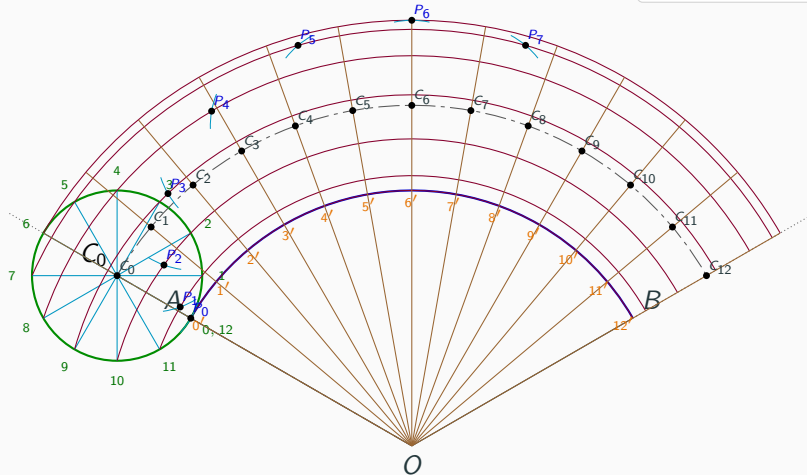


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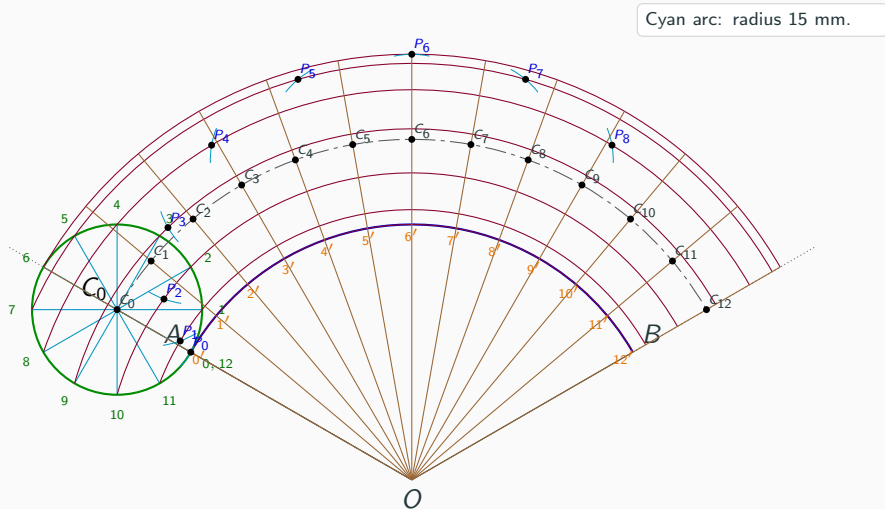
Cyan arc: radius 15 mm.

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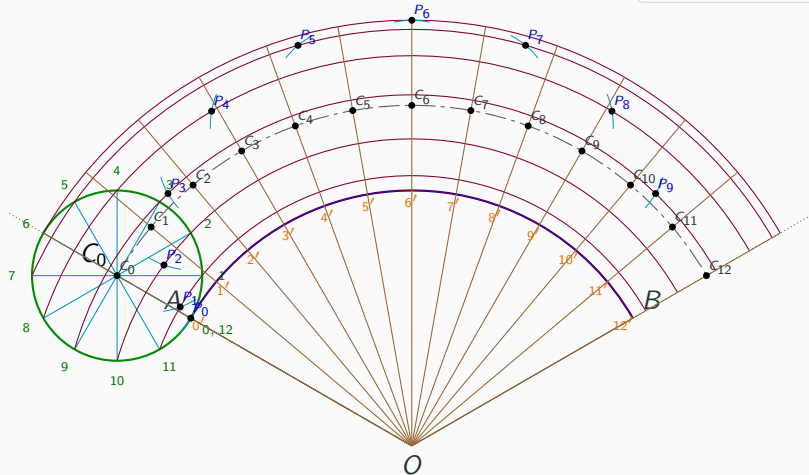


Cyan arc: radius 15 mm.

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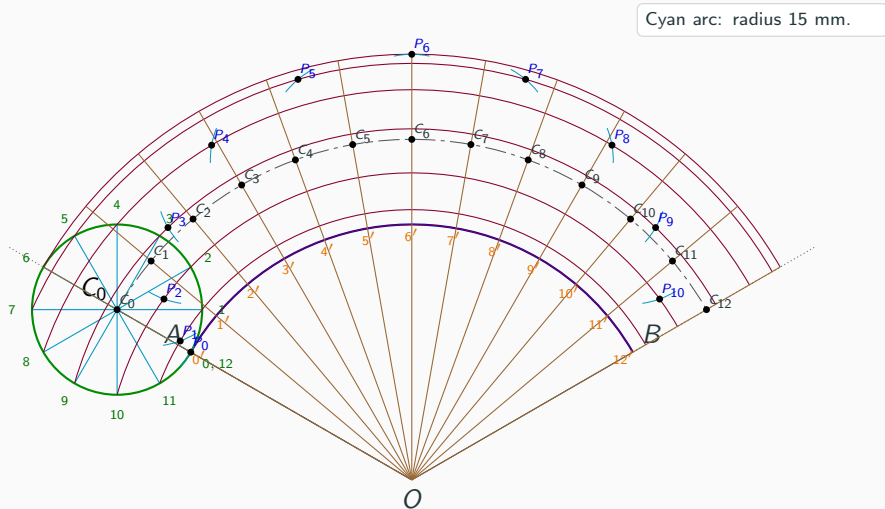


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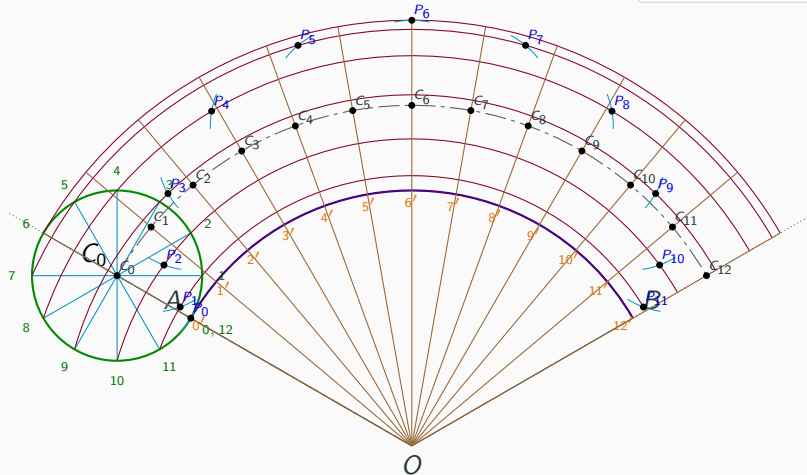
Cyan arc: radius 15 mm.

## Step 13: Draw the small cutting arc and locate $P_i$



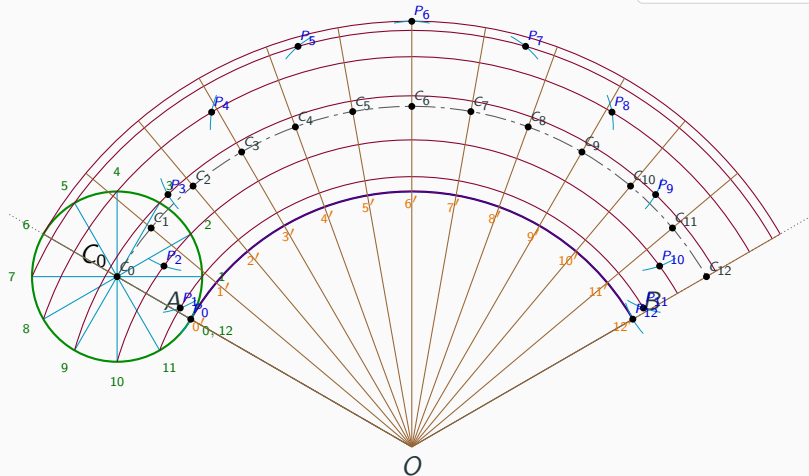
### Step 13: Draw the small cutting arc and locate $P_i$

Cyan arc: radius 15 mm.

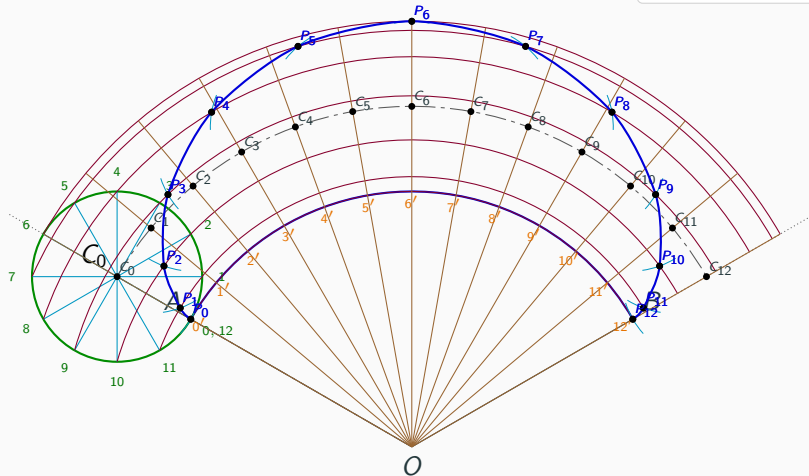




## Step 13: Draw the small cutting arc and locate $P_i$

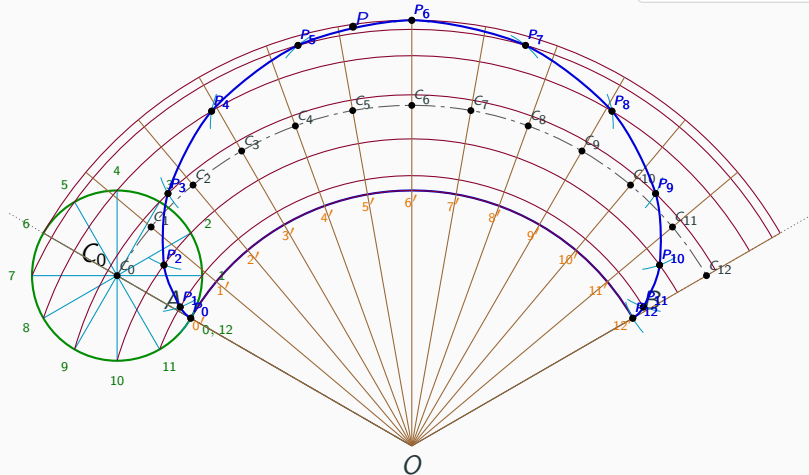


## Step 14: Draw the epicycloid



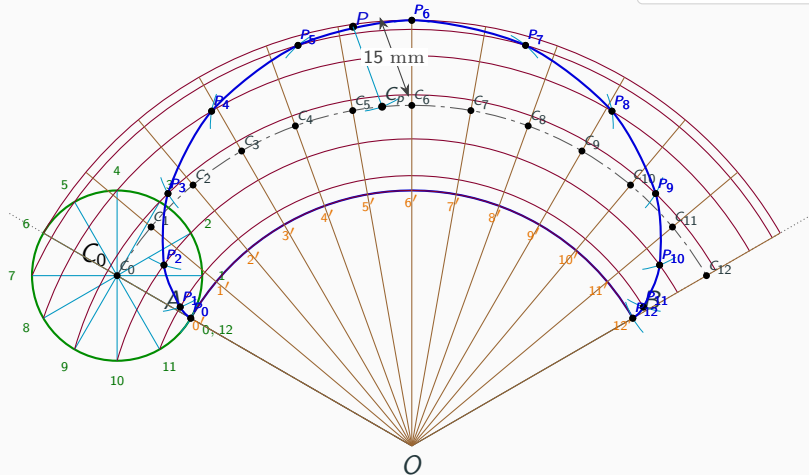
Blue curve: epicycloid.

## Step 15: Select a point $P$ on the epicycloid

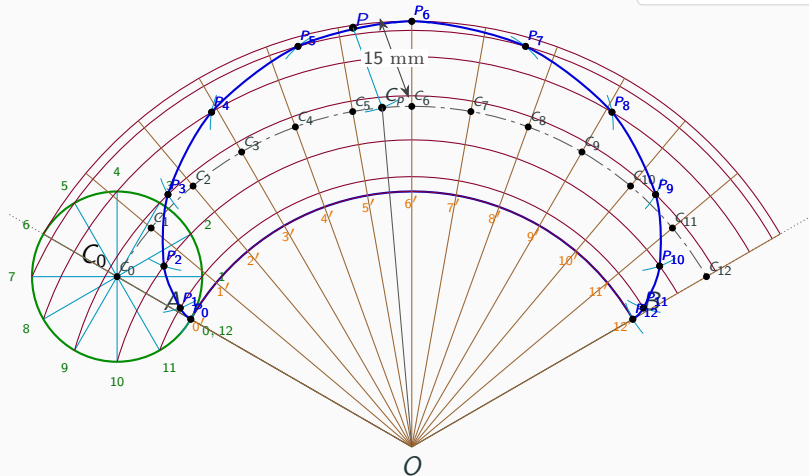


## Step 16: Draw the 15 mm arc from $P$ to the centre curve

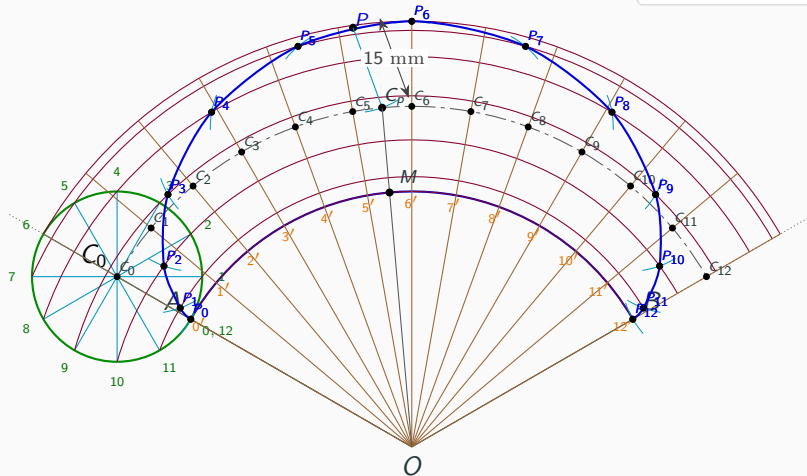
$$PC_P = 15 \text{ mm}$$



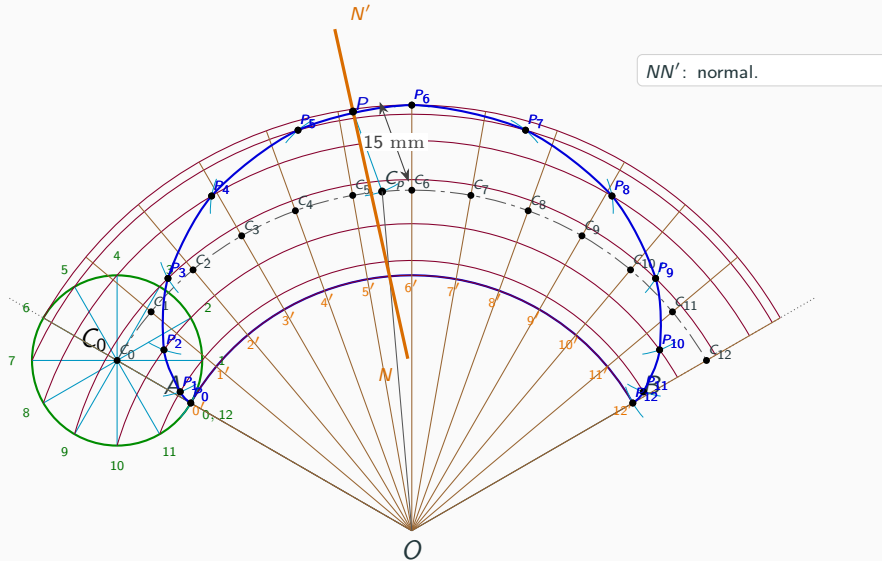
## Step 17: Join $C_p$ to $O$



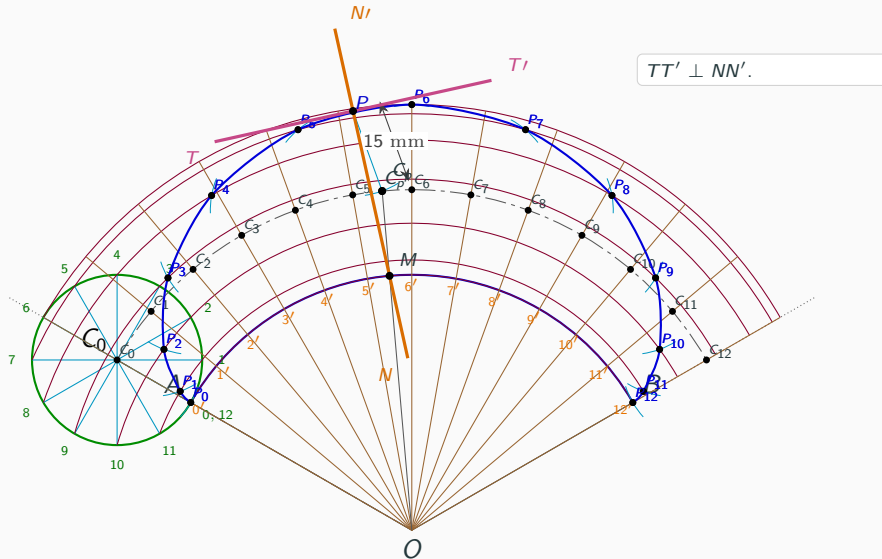
## Step 18: Mark $M$ on the directing circle



## Step 19: Draw the normal $NN'$ through $P$

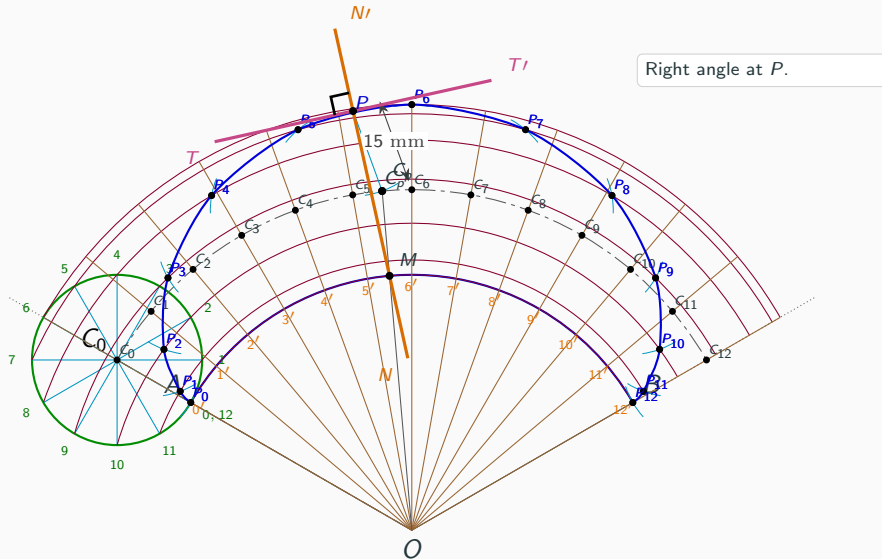


## Step 20: Draw the tangent perpendicular to $NN'$





## Step 20: Show the right-angle at $P$



## Step 20: Final completed epicycloid construction

